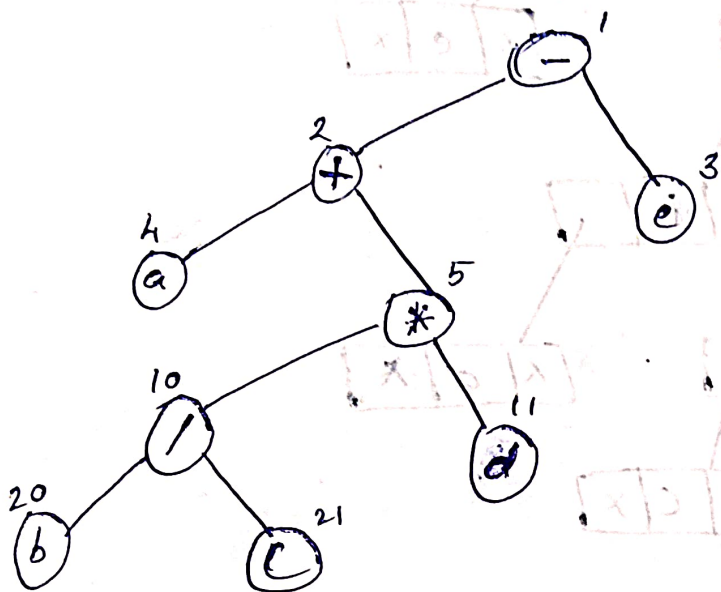


1) Construct a binary tree for the expression  
 $(a + ((b / c) * d)) - e$

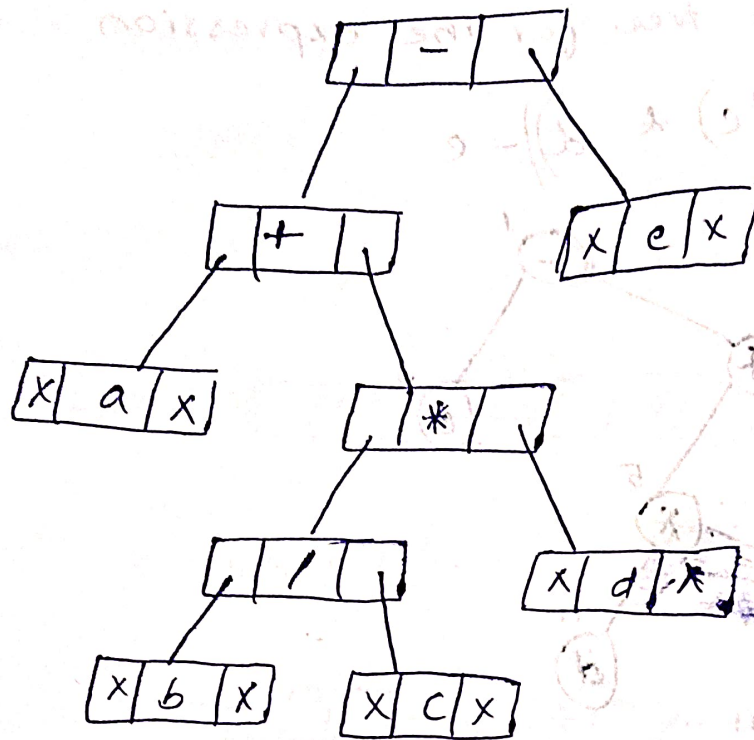


Post order :  $a \ b \ c \ / \ d \ * \ + \ e \ -$

2) Static representation of Binary tree

| index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | ... | 20 | 21 |
|-------|---|---|---|---|---|---|---|---|---|----|----|-----|----|----|
|       | - | + | e | a | * |   |   |   |   | /  | d  |     | b  | c  |

3) Dynamic representation of binary tree



### 3) Memory Representation of binary tree

| Index | Left | Right data | Right |
|-------|------|------------|-------|
| 100   | 200  | -          | 300   |
| 200   | 400  | +          | 300   |
| 300   | NULL | e          | NULL  |
| 400   | NULL | a          | NULL  |
| 500   | 600  | *          | 700   |
| 600   | 800  | /          | 900   |
| 700   | NULL | d          | NULL  |
| 800   | NULL | b          | NULL  |
| 900   | NULL | c          | NULL  |

#### H) Binary tree Traversal

Preorder (N-L-R)

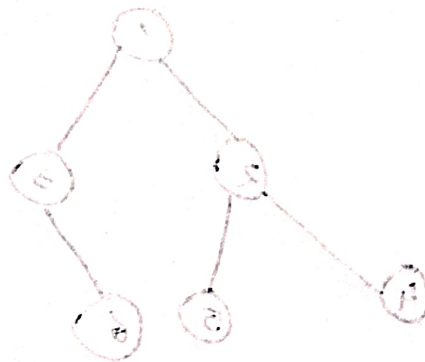
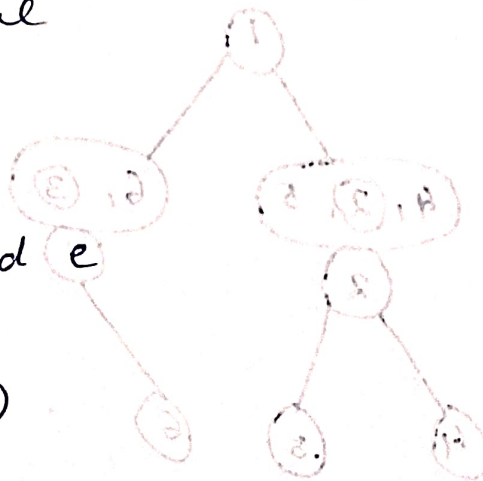
$- + a * / b c d e$

Inorder (L-N-R)

$a + b / c * d - e$

Postorder (L-R-N)

$a b c / d * + e -$



B) Inorder = 4 2 5 1 6 3

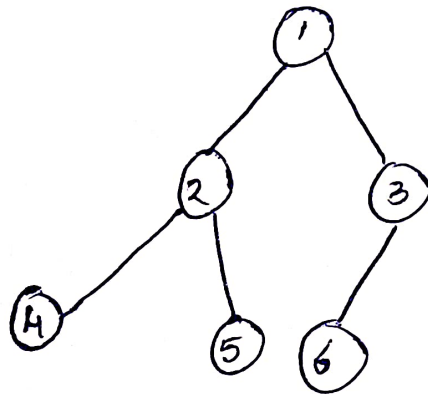
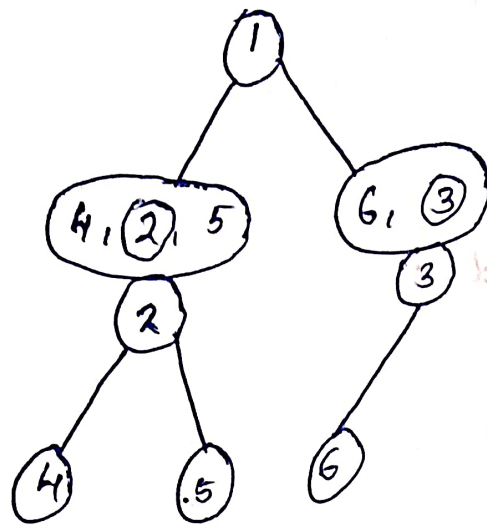
Postorder = 4 5 2 6 3 1 - last node is root node

In post order last node is the root node.  
while checking the inorder traversal we get that  
the elements to the left of root node is  
the left child and elements to the right is  
the right child.

Root = 1

Left child = 4 2 5

Right child = 6 3



Preorder Traversal - 1 2 4 5 3 6